

Saras Ugale

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Professional Summary

Results-driven AI/ML Engineer with hands-on experience designing, training, and deploying machine learning models across computer vision, NLP, and generative AI domains. Proficient in PyTorch, TensorFlow, Scikit-learn, and MLOps workflows, with a strong foundation in model fine-tuning, data preprocessing, and production-grade deployment using FastAPI and Flask. Demonstrated ability to deliver end-to-end ML pipelines achieving measurable performance improvements. Seeking an ML/AI Engineer role to build intelligent systems at scale.

Technical Skills

Languages & Frameworks:	Python, SQL, PyTorch, TensorFlow, Scikit-learn, Keras
ML/AI:	Supervised/Unsupervised Learning, CNNs, RNNs, LSTMs, Transformer Models
Generative AI & LLMs:	LangChain, Hugging Face, RAG, Fine-Tuning LLMs, Prompt Engineering
MLOps & Deployment:	Model Deployment, FastAPI, Flask, REST APIs, GitHub Actions
Data Science:	Feature Engineering, Data Preprocessing, Pandas, NumPy, Matplotlib, Seaborn
Tools & Platforms:	AWS, Google Cloud Vision (OCR), FAISS, Vector Databases, OpenCV, Git
Visualization:	Power BI, Tableau, Matplotlib

Project Work

MediCode AI - Explainable ICD-10 Medical Coding Assistant

[GitHub](#)

- Built an AI-powered medical coding assistant using BioBERT, PubMedBERT, FAISS, and KNN, achieving **95.10% Recall@5**, **93.80% MRR**, and **96.31% nDCG@5** on ICD-10 code retrieval tasks.
- Developed a scalable retrieval-based architecture using semantic search and vector similarity, eliminating the need for retraining when new ICD codes are introduced.
- Integrated Google Cloud Vision OCR and PyMuPDF to process handwritten prescriptions, scanned documents, and PDFs for automated clinical text extraction.
- Implemented an Explainable AI (XAI) layer with LLM-generated reasoning, keyword highlighting, and confidence-based recommendations through a FastAPI dashboard.

DeepShield AI - Deepfake Video Detection System

[GitHub](#)

- Developed a deepfake detection system using a **ResNeXt-CNN + LSTM** architecture for spatial-temporal analysis of video sequences, achieving **87% classification accuracy** on the **Celeb-DF v2** dataset.
- Built a multi-stage preprocessing pipeline including face extraction, frame sampling, normalization, and sequence generation using OpenCV.
- Benchmarked CNN-only and CNN+LSTM models, validating the effectiveness of temporal modeling for detecting subtle manipulation artifacts.
- Deployed an end-to-end Flask web application for video upload, automated frame analysis, and real-time authenticity prediction.

Education

Bachelor of Engineering, Computer Science and Engineering (Artificial Intelligence) Jan 2022 - Jul 2026
Nutan College of Engineering & Research, Pune | DBATU

Certifications

- AWS Machine Learning Learning Plan**- AWS Skill Builder
- Deep Learning for Beginners** - Simplilearn
- AI Fundamentals: Language and Vision in AI** - IBM
- British Airways- Data Science Job Simulation** - Forage
- AI Fluency Framework Foundations certificate** - Anthropic
- Data Science & Analytics** - HP LIFE